

IIE8560 HW02 - 장서현 2025324096

1. Consider the joint distribution $p(X, Y)$

		x				
		1	2	3	4	$p(y)$
y	1	1/8	1/16	1/32	1/32	8/32
	2	1/16	1/8	1/32	1/32	8/32
	3	1/16	1/16	1/16	1/16	8/32
	4	1/4	0	0	0	8/32
$p(x)$		1/2	1/4	1/8	1/8	

a. What is the joint entropy $H(X, Y)$? 3.375 bits

b. What are the marginal entropies $H(X)$ and $H(Y)$?

1.75 bits 2 bits

c. The entropy of X conditioned on a specific value of y is defined as

$$H(X|Y=y) = - \sum_x p(x|y) \log p(x|y)$$

Compute $H(X|y)$ for each value of y . Does the posterior entropy on X ever increase

given an observation of Y ? 1.75, 1.75, 2, 0 Yes. $H(X|Y=3) > H(X)$

d. The conditional entropy is defined as

$$H(X|Y) = - \sum_y p(y) H(X|Y=y)$$

Compute this. Does the posterior entropy on X increase or decrease when averaged over the possible values of Y ? 1.375 bits, Decrease.

e. What is the mutual information between X and Y ? 0.375 bits

$$\begin{aligned} a. H(X, Y) &= - \sum_x \sum_y p(x, y) \log_2 p(x, y) \\ &= - [4/32 \log_2 (4/32) + 2/32 \log_2 (2/32) + 1/32 \log_2 (1/32) + 1/32 \log_2 (1/32) \\ &\quad + 2/32 \log_2 (2/32) + 4/32 \log_2 (4/32) + 1/32 \log_2 (1/32) + 1/32 \log_2 (1/32) \\ &\quad + 2/32 \log_2 (2/32) + 2/32 \log_2 (2/32) + 7/32 \log_2 (7/32) + 2/32 \log_2 (2/32) \\ &\quad + 8/32 \log_2 (8/32)] = 3.375 \text{ bits} \end{aligned}$$

$$b. H(X) = - [1/2 \log_2 (1/2) + 1/4 \log_2 (1/4) + 1/8 \log_2 (1/8) + 1/8 \log_2 (1/8)] = 1.75 \text{ bits}$$

$$H(Y) = - [1/4 \log_2 (1/4) + 1/4 \log_2 (1/4) + 1/4 \log_2 (1/4) + 1/4 \log_2 (1/4)] = 2 \text{ bits}$$

$$c. H(X|Y=1) = - [1/2 \log_2 (1/2) + 1/4 \log_2 (1/4) + 1/8 \log_2 (1/8) + 1/8 \log_2 (1/8)] = 1.75 \text{ bits}$$

$$H(X|Y=2) = - [1/4 \log_2 (1/4) + 1/2 \log_2 (1/2) + 1/8 \log_2 (1/8) + 1/8 \log_2 (1/8)] = 1.75 \text{ bits}$$

$$H(X|Y=3) = - [4 \times 1/4 \log_2 (1/4)] = 2 \text{ bits} \rightarrow H(X|Y=3) = 2 > H(X) = 1.75$$

$$H(X|Y=4) = - [-1 \log_2 (1)] = 0 \text{ bits}$$

$$d. H(X|Y) = 1/4 \times 1.75 + 1/4 \times 1.75 + 1/4 \times 2 + 1/4 \times 0 = 1.375 \text{ bits}$$

$$\rightarrow H(X|Y) = 1.375 < H(X) = 1.75$$

$$e. I(X; Y) = H(X) - H(X|Y) = 1.75 - 1.375 = 0.375 \text{ bits}$$

2. Let p and q be pmfs on a finite set $\mathcal{X}$. Assume $q(x) > 0$ whenever $p(x) > 0$.

Define

$$D(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} \text{ (any log base).}$$

a. Show that $D(p \parallel q) \geq 0$. You can apply the log-sum inequality to $\{p(x)\}, \{q(x)\}$ (below)

b. State when equality $D(p \parallel q) = 0$ holds. iff $p(x) = q(x)$ for all $x \in \mathcal{X}$

a. Since p & q are pmfs and since we assume $q(x) > 0$ whenever $p(x) > 0$, $\frac{p(x)}{q(x)}$ is well-defined.

Applying the Log-sum inequality to the definition of KL divergence,

$$D(p \parallel q) = \sum_{x \in \mathcal{X}} p(x) \log_b \frac{p(x)}{q(x)} \geq \left(\sum_{x \in \mathcal{X}} p(x) \right) \log_b \frac{\sum_{x \in \mathcal{X}} p(x)}{\sum_{x \in \mathcal{X}} q(x)}$$

$p(x)$ and $q(x)$ being pmfs, $\sum_{x \in \mathcal{X}} p(x) = 1$ and $\sum_{x \in \mathcal{X}} q(x) = 1$

Thus, $D(p \parallel q) \geq (1) \log_b \frac{1}{1} = 0$.

b. Equality in Log-sum inequality holds if and only if $\frac{a_x}{b_x} = c$ for some constant c , for all x where $a_x > 0$.

Thus equality $D(p \parallel q) = 0$ holds if and only if $\frac{p(x)}{q(x)} = c$ for all x such that $p(x) > 0$.

This implies $p(x) = c \cdot q(x)$ for all x where $p(x) > 0$.

$$\sum_{x \in \mathcal{X}} p(x) = \sum_{x \in \mathcal{X}} c \cdot q(x) = 1$$

$$\rightarrow c = 1$$

Thus, equality holds if and only if $p(x) = q(x)$ for all x where $p(x) > 0$ (for all $x \in \mathcal{X}$)

3. You are building a binary classifier with posterior score as $\eta(x) = \Pr(y = 1 | x)$. The decision costs are:

- false positive (predict 1 when $y = 0$): c_{FP}
- false negative (predict 0 when $y = 1$): c_{FN}

a. Derive the Bayes-optimal rule in terms of a threshold on $\eta(x)$. $\eta(x) \geq \frac{c_{FP}}{c_{FN} + c_{FP}}$

b. Evaluate the numeric threshold when $c_{FP} = 2$ and $c_{FN} = 1$. $\frac{2}{3}$

$$\begin{aligned} p(\hat{y}=0|x) &= l_{00} \cdot p(y=0|x) + l_{01} \cdot p(y=1|x) \\ &= 0 \cdot (1-\eta(x)) + c_{FN} \cdot \eta(x) = c_{FN} \cdot \eta(x) \end{aligned}$$

$$\begin{aligned} p(\hat{y}=1|x) &= l_{10} \cdot p(y=0|x) + l_{11} \cdot p(y=1|x) \\ &= c_{FP} \cdot (1-\eta(x)) + 0 \cdot \eta(x) = c_{FP} \cdot (1-\eta(x)) \end{aligned}$$

$$\begin{aligned} \text{Predict } \hat{y}=1 \text{ if } p(\hat{y}=1|x) &\leq p(\hat{y}=0|x) \\ c_{FP}(1-\eta(x)) &\leq c_{FN} \cdot \eta(x) \\ \eta(x) &\geq \frac{c_{FP}}{c_{FN} + c_{FP}} \end{aligned}$$

$$b. \frac{c_{FP}}{c_{FN} + c_{FP}} = \frac{2}{1+2} = \frac{2}{3}$$

4. You fit two linear regression models on the same dataset of size $N = 80$.

- M_1 : $D_1 = 3$, $RSS_1 = 220$.
- M_2 : $D_2 = 5$, $RSS_2 = 200$.

Assume Gaussian noise with common variance and compare models using:

$$AIC(M) = N \log\left(\frac{RSS}{N}\right) + 2D,$$

$$BIC(M) = N \log\left(\frac{RSS}{N}\right) + D \log N$$

a. Report $AIC(M_1)$, $AIC(M_2)$ and state which model AIC prefers.

$$86.93 \quad 83.3 \quad M_2$$

b. Compute $BIC(M_1)$, $BIC(M_2)$ and state which model BIC prefers.

$$94.07 \quad 95.21 \quad M_1$$

$$a. AIC(M_1) = 80 \log\left(\frac{220}{80}\right) + 2(3) = 86.93$$

$$AIC(M_2) = 80 \log\left(\frac{200}{80}\right) + 2(5) = 83.3 \rightarrow \text{AIC prefers } M_2$$

$$b. BIC(M_1) = 80 \log\left(\frac{220}{80}\right) + 3 \log(80) = 94.07 \rightarrow \text{BIC prefers } M_1$$

$$BIC(M_2) = 80 \log\left(\frac{200}{80}\right) + 5 \log(80) = 95.21$$